Welcome

CS 472/572
Unsupervised Machine Learning

PCA Analysis

Week-9 -Class -1

Fall 2025

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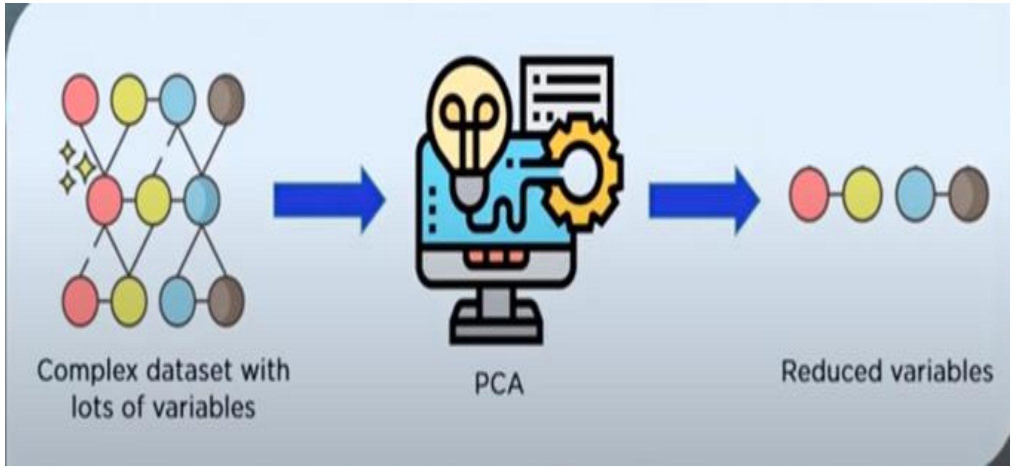
Topics

1. Eigenvectors and eigenvalues
2. Variance and multivariate Gaussian distributions
3. Computing a covariance matrix from data
4. Principal Components Analysis (PCA)

Introduction

Principal Component Analysis, commonly referred to as PCA, is a powerful mathematical technique used in data analysis and statistics. At its core, PCA is designed to simplify complex datasets by transforming them into a more manageable form while retaining the most critical information.

- reducing the dimensionality of dataset
- Increasing interpretability without losing information



Dimensionality Reduction

Dimensionality reduction refers to the techniques that reduce the number of input variables in a dataset.

Why DR?

- Less dimensions for a given dataset means less computation or training time
- Redundancy is removed after removing similar entries from the dataset
- Data Compression (Reduce storage space)
- It helps to find out the most significant features and skip the rest
- Leads to better human interpretations

Why PCA?

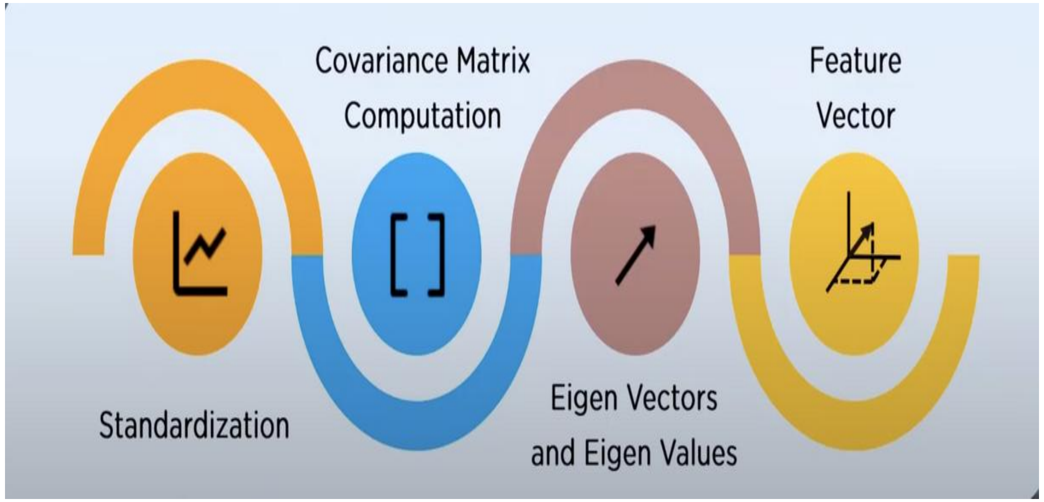
- Dimensionality Reduction
- Noise Reduction
- Visualization
- Feature Engineering
- Overfitting Problem
- Data Compression
- Machine Learning Processing

Important Terminologies

- Variance
- Covariance
- Eigenvalues
- Eigenvectors
- Principle Component

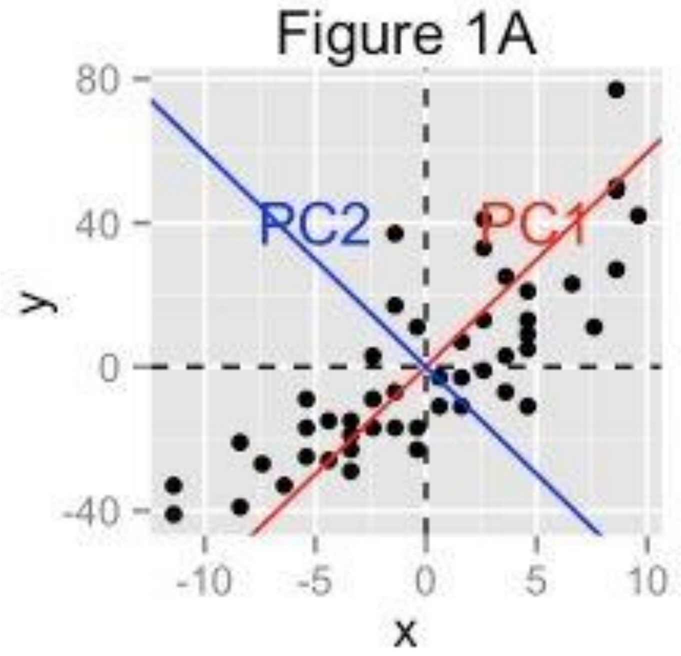
How does PCA work?

Step 1: Standardize the data.
Step 2: Calculate the covariance matrix.
Step 3: Compute the eigenvectors and eigenvalues.
Step 4: Select the principal components.
Step 5: Project data onto the new basis.



Principal Component Analysis

- A mathematical procedure with ***simple matrix operations*** from ***linear algebra and statistics*** to transform a number of *correlated variables into smaller number of uncorrelated variables* called principal components.
- Emphasize variation and bring out strong patterns in a dataset.
  - To make data easy to ***explore and visualize***.
- Still contains most of the information in the large set.



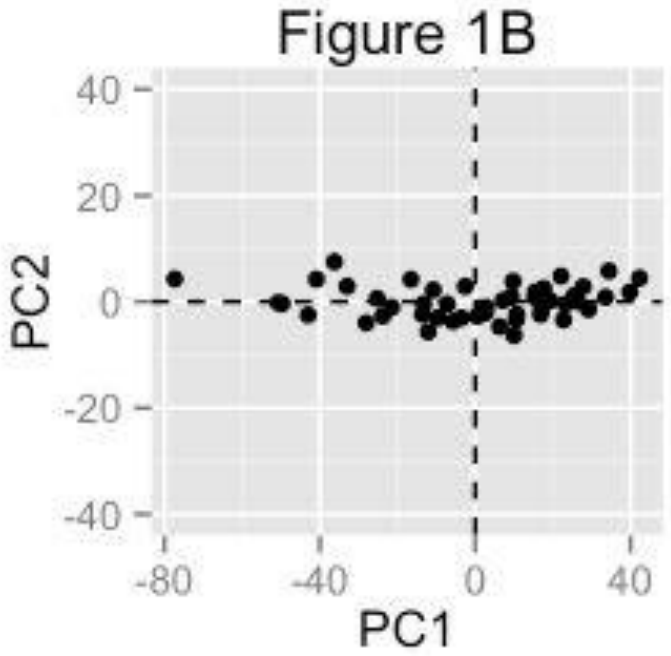


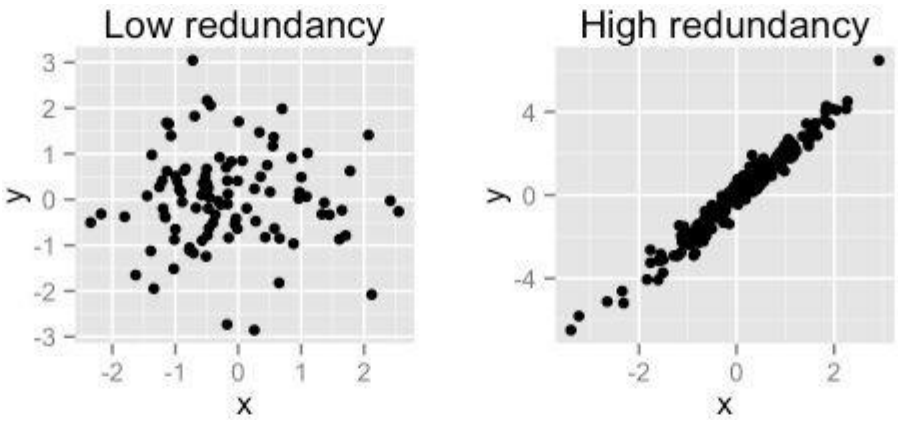
Figure 1A: The data are represented in the X-Y coordinate system

Figure 1B: The *PC1 axis* is the **first principal direction** along which the samples show the largest variation

Goals of PCA

- To ***identify hidden pattern*** in a data set
- To ***reduce the dimensionality*** of the data by removing the noise and redundancy in the data
- To identify ***correlated variables***

- PCA method is particularly useful when the variables within the data set are highly correlated.
- **Correlation** indicates that there is **redundancy** in the data.
- PCA can be used to reduce the original variables into a smaller number of new variables (= **principal components**) explaining most of the variance in the original variables.



Steps to implement PCA in 2D dataset

Step 1: **Normalize** the data

Step 2: Calculate the **covariance matrix**

Step 3: Calculate the **eigenvalues and eigenvectors**

Step 4: **Choosing** principal components

Step 5: Forming a **feature vector**

Step 6: **Forming Principal Components**

Step 1: Normalization

This is done by subtracting the respective means from the numbers in the respective column. So if we have two dimensions X and Y, all X become x- and all Y become y-.

For all X; $x- = X - \mu_x$

For all Y; $y- = Y - \mu_y$

This produces a dataset whose mean is zero.

Step 2: Calculation of correlation

$$\text{Matrix (covariance)} = \begin{bmatrix} var(x) & var(x,y) \\ var(y,x) & var(y) \end{bmatrix}$$

Covariance Matrix for Iris Dataset

	Sepal.Length	*Sepal.Width*	*Petal.Length*	*Petal.Width*
Sepal.Length	0.69	-0.04	1.27	0.52
Sepal.Width	-0.04	0.19	-0.33	-0.12
Petal.Length	1.27	-0.33	3.12	1.30
Petal.Width	0.52	-0.12	1.30	0.58

Calculation of correlation(contd...)

If **x** and **y** be two variables with length n,

The variance of **x** , variance of **y** and variance of **x & y** is given by following equations.

$\boldsymbol{m_x}$: mean of **x** variables

$\boldsymbol{m_y}$: mean of **y** variables

$$\sigma_{xx}^2 = \frac{\sum_i (x_i - m_x)(x_i - m_x)}{n-1}$$

$$\sigma_{yy}^2 = \frac{\sum_i (y_i - m_y)(y_i - m_y)}{n-1}$$

$$\sigma_{xy}^2 = \frac{\sum_i (x_i - m_x)(y_i - m_y)}{n-1}$$

Correlation is the index to measure how strongly two variable are related to each other. The value of the same ranges for ***-1*** to ***+1***. (i.e. $-1 < r < 1$)

If $r<0$, variables are ***negatively correlated*** (e.g. x increases when y increases)

If $r>0$, variables are ***positively correlated*** (e.g. x increases when y decreases)

If $r=0$, variables has ***no correlation***

Step 3: Calculation of Eigenvalue and Eigenvector

Calculate Eigenvalue and Eigenvector of the covariance matrix using power method:

$$|\lambda - AI| = 0$$

Where, I is an identity matrix of same dimension as A λ is eigenvalue.

For each value of λ, corresponding eigenvector 'v' is obtained by solving:

$$(\lambda - AI)v = 0$$

Step 4: Choosing Component

- Eigenvalues from largest to smallest so that it gives us the components in order or significance.
- If we have a dataset with *n* variables, then we have the corresponding *n* eigenvalues and eigenvectors.
- Eigenvector (v) corresponding to largest eigenvalue (λ) is called first principal component.
- To reduce the dimensions, we choose the first *p* eigenvalues and ignore the rest.

Step 5: Forming a feature vector

Form a feature vector consists of selected eigenvectors

Feature Vector = (eig1, eig2)

NewData = FeatureVector^T x ScaledData^T

NewData is the Matrix consisting of the principal components,

FeatureVector is the matrix we formed using the eigenvectors we chose to keep, and

ScaledData is the scaled version of original dataset

Sample Implementation with sklearn

```
# Principal Component Analysis
from numpy import array
from sklearn.decomposition import PCA
# define a matrix
A = array([[1, 2], [3, 4], [5, 6]])
print(A)
# create the PCA instance
pca = PCA(2)
# fit on data
pca.fit(A)
# access values and vectors ie.covariance matrix
print(pca.components_)
print(pca.explained_variance_)
# transform data
B = pca.transform(A)
print(B)
```

Output

```
[[1 2][3 4][5 6]]

[[ 0.70710678  0.70710678]
 [ 0.70710678 -0.70710678]]

[  8.00000000e+00   2.25080839e-33]

[[ -2.82842712e+00   2.22044605e-16]
 [  0.00000000e+00   0.00000000e+00]
 [  2.82842712e+00
-2.22044605e-16]]
```

Applications of PCA

- **Commonly used dimensionality reduction technique in domains like**

facial recognition*, *computer vision* and *image compression.

- Used in ***finding patterns in data of high dimension*** in the field of

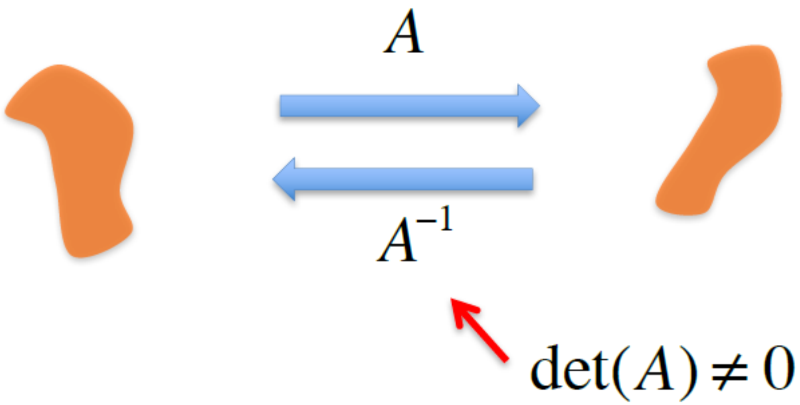
***finance*, *data mining*, *bioinformatics*, *psychology*.**

Eigenvectors and eigenvalues

Matrix transformations

$$\vec{y} = A\vec{x}$$

- In general A maps the set of vectors in $\mathbb{R}^2$ onto another set of vectors in $\mathbb{R}^2$.

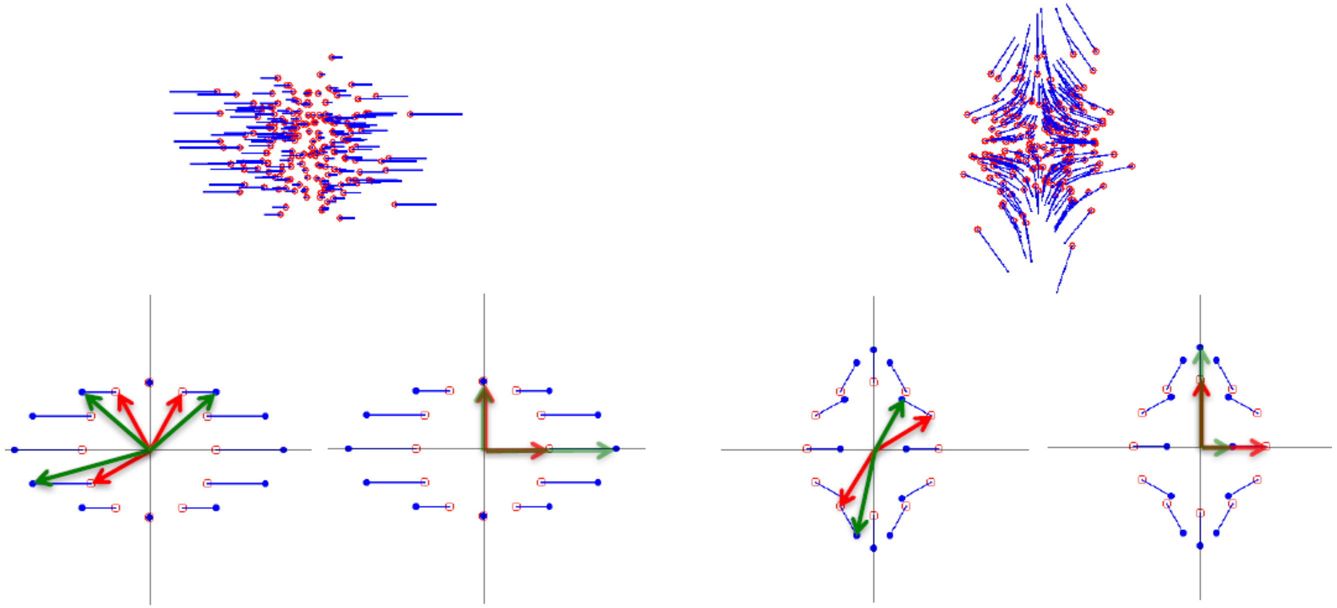


Eigenvectors and eigenvalues

- Matrix transformations have special directions

$$A = \begin{pmatrix} 1+\delta & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1-\delta & 0 \\ 0 & 1+\delta \end{pmatrix}$$

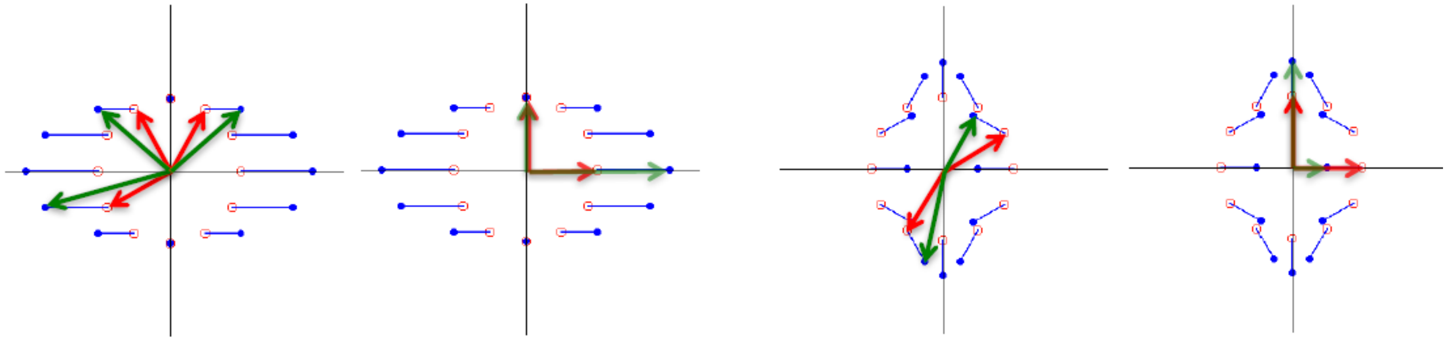


These are all diagonal matrices

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Eigenvectors and eigenvalues

- Some vectors are rotated, some are not.



- For a diagonal matrix, vectors along the axes are scaled, but not rotated.

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \Lambda \hat{e}_1 = \lambda_1 \hat{e}_1$$

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \lambda_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \Lambda \hat{e}_2 = \lambda_2 \hat{e}_2$$

Eigenvectors and eigenvalues

- Diagonal matrices have the property that they map any vector parallel to a standard basis vector into another vector along that standard basis vector.

eigenvalue equation

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \qquad \Lambda \hat{e}_i = \lambda_i \hat{e}_i \;\;, \;\; i = 1,2\ldots.n$$

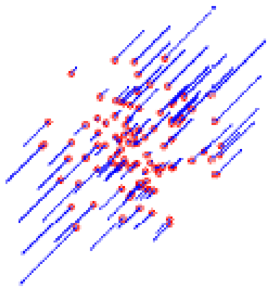
- Any vector $\vec{v}$ that is mapped by matrix A onto a parallel vector $\lambda\vec{v}$ is called an eigenvector of A. The scale factor λ is called the eigenvalue of vector $\vec{v}$.

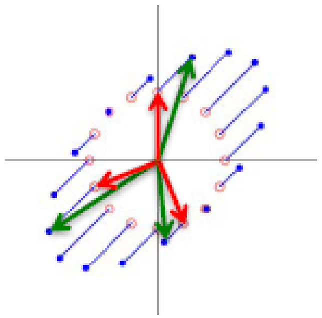
- A matrix in $\mathbb{R}^n$ has n eigenvectors and n eigenvalues.

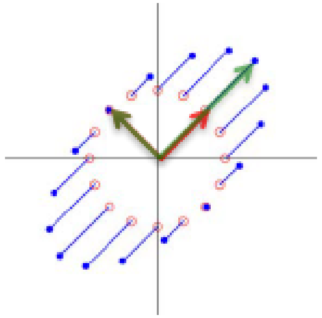
Eigenvectors and eigenvalues

- What are the special directions of our rotated transformations?

$$A = \Phi\Lambda\Phi^T = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix} \qquad \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \qquad \Phi(45^0) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$







Eigenvectors and eigenvalues

What are the eigenvectors and eigenvalues of our rotated transformation matrix

$$A\vec{x}_i = a_i \vec{x}_i$$

$$\Phi\Lambda\Phi^T \vec{x}_i = a_i \vec{x}_i$$

$$\Phi^T\Phi\Lambda\Phi^T \vec{x}_i = \Phi^T(a_i \vec{x}_i)$$

$$\Lambda\underbrace{\Phi^T\vec{x}_i}_{\hat{e}_i} = a_i \underbrace{\Phi^T\vec{x}_i}_{\hat{e}_i}$$

Remember…

$$\Lambda\hat{e}_i = \lambda_i \hat{e}_i \qquad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

So we know the solution if $\Phi^T\vec{x}_i = \hat{e}_i$

$$\Lambda\hat{e}_i = a_i \hat{e}_i$$

$$\Rightarrow a_i = \lambda_i$$

So the solution to the eigenvalue equation $\Phi\Lambda\Phi^T \vec{x}_i = a_i \vec{x}_i$ is:

eigenvalues

$$a_i = \lambda_i$$

eigenvectors

$$x_i = \theta e\hat{}_i$$

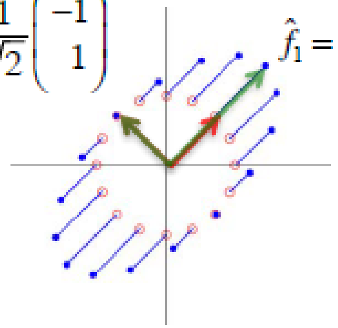
what is this?

Eigenvectors and eigenvalues

- The eigenvectors are just the standard basis vectors rotated by the matrix Φ !

$$\vec{x}_i = \Phi \hat{e}_i$$

$$\hat{f}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



$$\hat{f}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Phi(45^0) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\vec{x}_1 = \Phi \hat{e}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\vec{x}_2 = \Phi \hat{e}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The eigenvectors are just the columns of Φ!

Eigenvectors and eigenvalues

- In summary, a symmetric matrix A can always be written as follows:

$$A = \Phi \Lambda \Phi^T$$

where Φ is a rotation matrix and Λ is a diagonal matrix

- The eigenvectors of A are the columns of Φ (the basis vectors, $\hat{f}_i$)

$$\Phi = \left(\hat{f}_1 \,\middle|\, \hat{f}_2 \right)$$

- The eigenvalue associated with each eigenvector $\hat{f}_i$ is the diagonal element λ_i of Λ.

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Eigen-decomposition

- The process of writing a matrix A as $A = \Phi\Lambda\Phi^T$ is called eigen-decomposition. It works for any symmetric matrix.
  - the eigenvalues λ_i are real numbers
  - the eigenvectors $\hat{f}_i$ form an orthogonal basis set
- Let's rewrite…

$$A = \Phi\Lambda\Phi^T \qquad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$A\Phi = \Phi\Lambda\Phi^T\Phi$$

- Eigenvalue equation

$$A\Phi = \Phi\Lambda$$

is equivalent to the set of equations $A\,\hat{f}_i = \lambda_i\,\hat{f}_i$

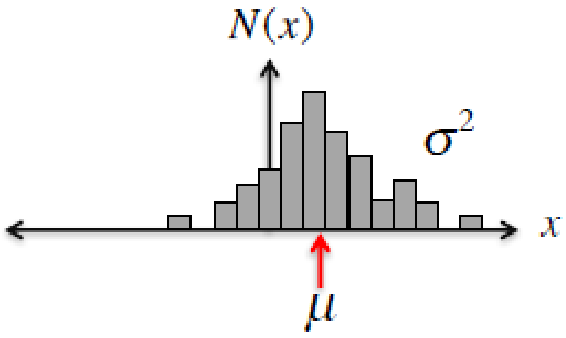
Variance and multivariate Gaussian distributions

Variance

- Let's say we have m observations of a variable x:

$$x^{(j)}\ ,\ j = 1,2,3....,m$$

j^{th} observation of x



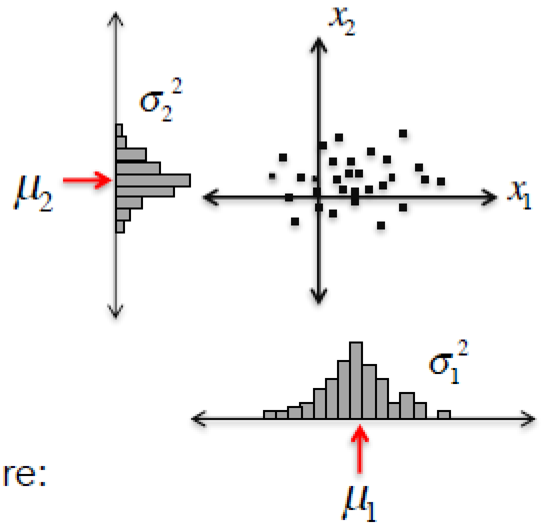
- The mean of these observations is $\mu = \langle x \rangle = \frac{1}{m}\sum_{i=1}^{m} x^{(i)}$

- The variance is $\sigma^2 = \frac{1}{m}\sum_{i=1}^{m}\left(x^{(i)} - \mu\right)^2$

Variance

- Now let's say we have m simultaneous observations of variables x_1 and x_2.

$$\begin{pmatrix} x_1^{(j)} \\ x_2^{(j)} \end{pmatrix}, \quad j = 1,2,3\ldots,m$$

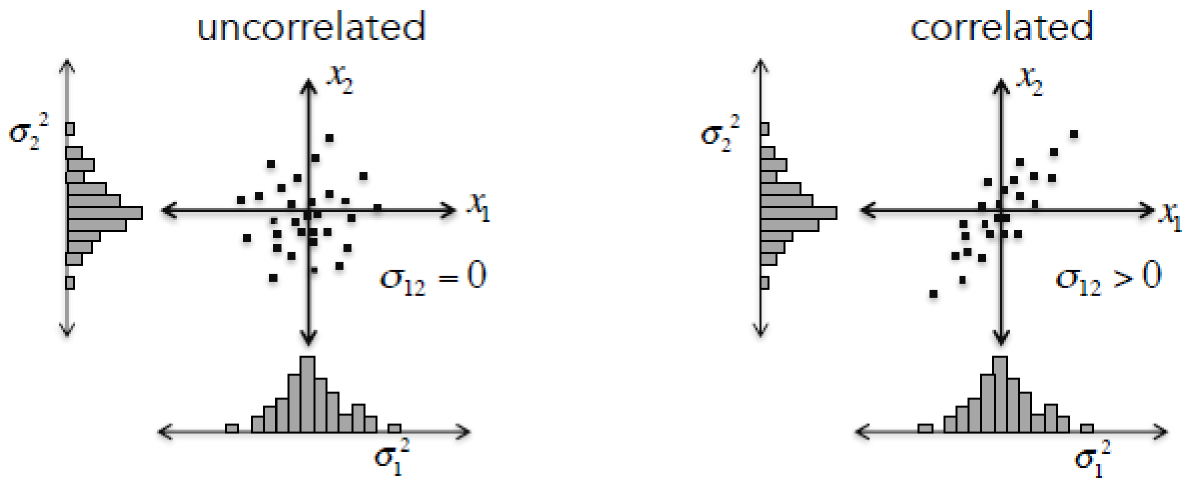


- The mean and variance of x_1 and x_2 are:

$$\mu_1 = \frac{1}{m}\sum_{j=1}^{m} x_1^{(j)} \qquad \mu_2 = \frac{1}{m}\sum_{j=1}^{m} x_2^{(j)}$$

$$\sigma_1^2 = \frac{1}{m}\sum_{j=1}^{m}\left(x_1^{(j)} - \mu_1\right)^2 \qquad \sigma_2^2 = \frac{1}{m}\sum_{j=1}^{m}\left(x_2^{(j)} - \mu_2\right)^2$$

Covariance



- x_1 has the same variance in both of these cases… also x_2
- So we need another measure to describe the relation between x_1 and x_2

covariance

$$\sigma_{12} = \frac{1}{m}\sum_{j=1}^{m}\left(x_1^{(j)} - \mu_1\right)\left(x_2^{(j)} - \mu_2\right)$$

correlation

$$\rho = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

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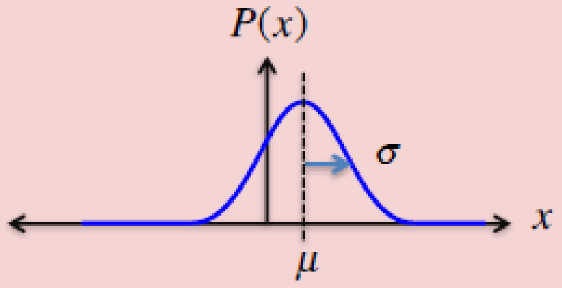
Gaussian distribution

- Many kinds of data can be fit by a Gaussian distribution

If x is a Gaussian random variable



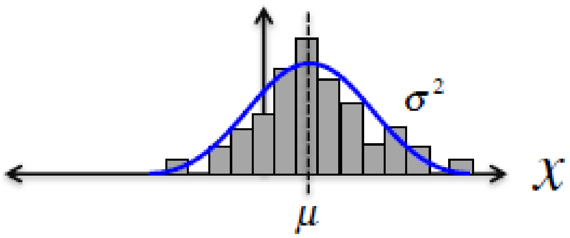
probability 'density'

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

- Gaussian is defined by only its mean and variance.

To find the Gaussian that best fits our data –

Just measure the mean and variance!



Multivariate Gaussian distribution

- We can create a Gaussian distribution in two dimensions

Two independent Gaussian random variables x_1 and x_2

$$P(x_1, x_2) = P(x_1)P(x_2)$$

$$= \beta e^{-\frac{1}{2}x_1^2} e^{-\frac{1}{2}x_2^2}$$

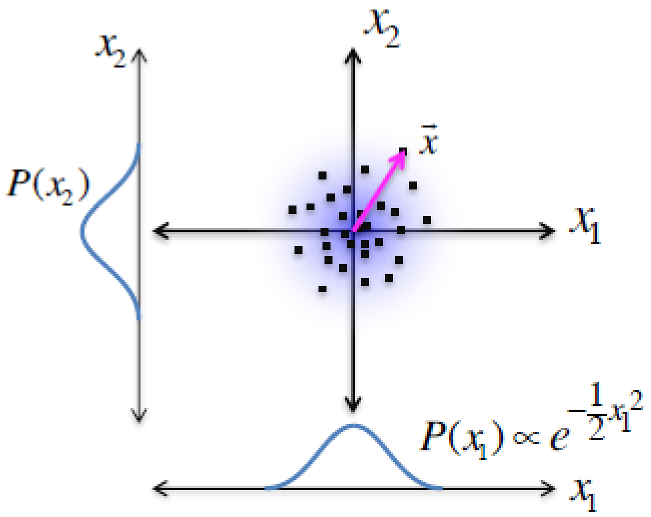
normalization to
make total prob = 1

$$= \beta e^{-\frac{1}{2}(x_1^2 + x_2^2)}$$

$$P(\vec{x}) = \beta e^{-\frac{1}{2}d^2}$$

$$d^2 = |\vec{x}|^2 = \vec{x}^T \vec{x}$$

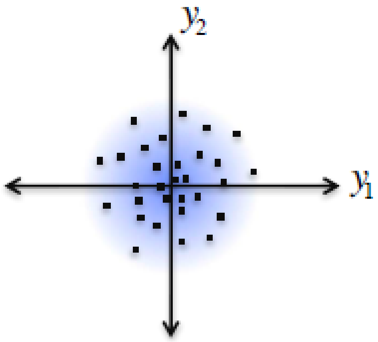
$$d = \text{Mahalanobis Distance}$$



Isotropic multivariate Gaussian distribution

2!

Multivariate Gaussian distribution

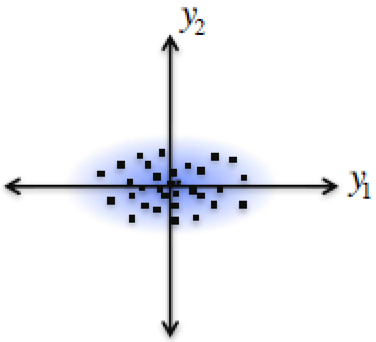


Isotropic

$$P(\vec{y}) = \beta e^{-\frac{1}{2\sigma^2}\vec{y}^T\vec{y}}$$

$$\sigma^2$$

variance

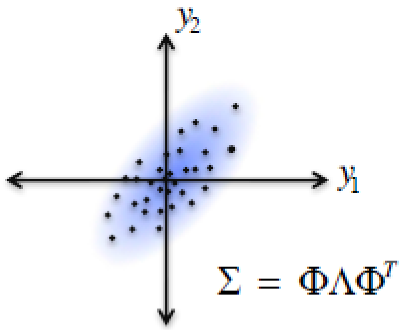


Non-isotropic:
no correlation

$$P(\vec{y}) = \beta e^{-\frac{1}{2}\vec{y}^T\Lambda^{-1}\vec{y}}$$

$$\Lambda = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}$$

variance matrix



Non-isotropic:
with correlation

$$P(\vec{y}) = \beta e^{-\frac{1}{2}\vec{y}^T\Sigma^{-1}\vec{y}}$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$$

covariance matrix

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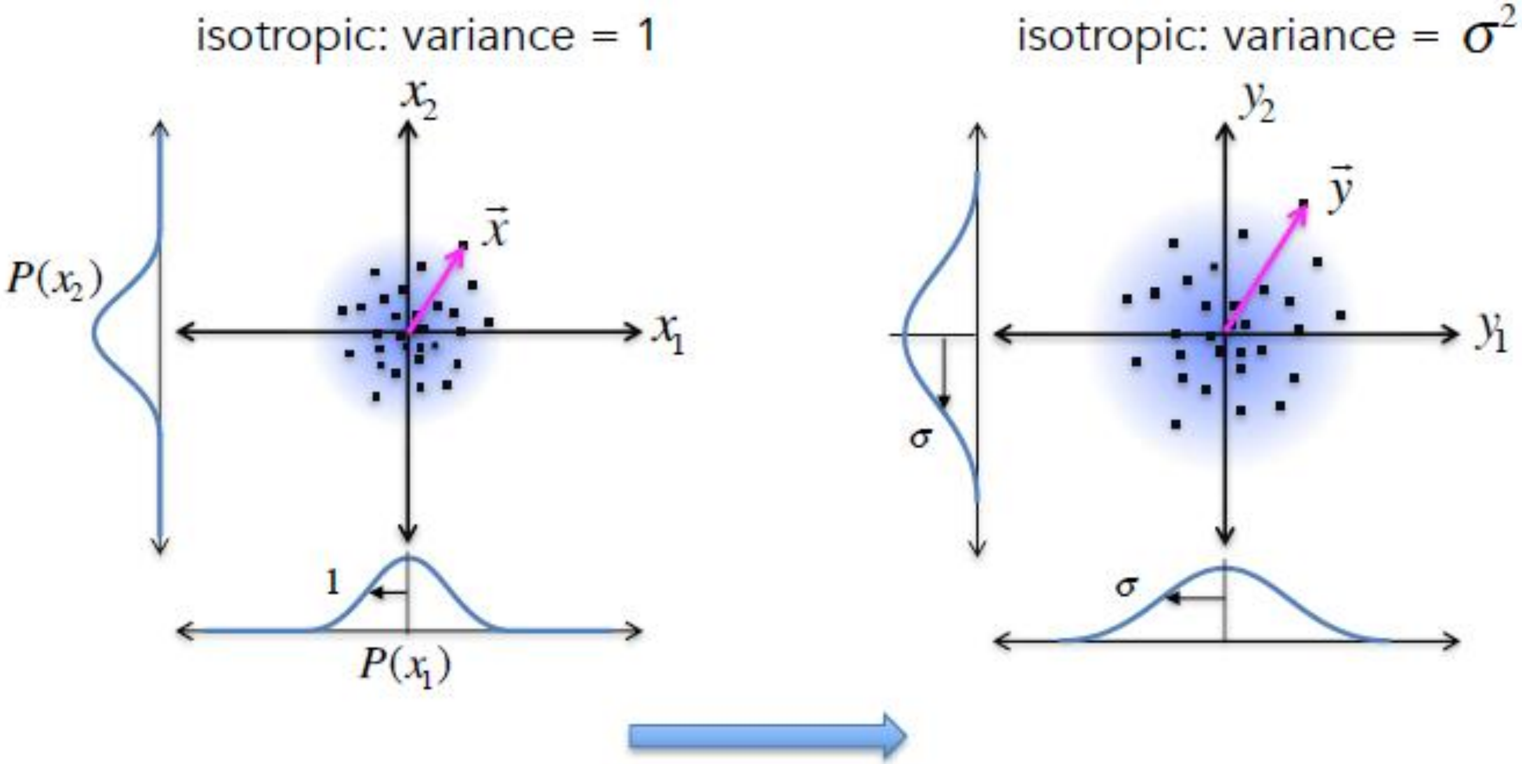
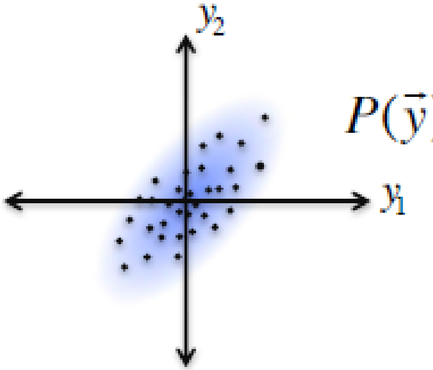
Multivariate Gaussian distribution



$$\vec{y} = \sigma \, \vec{x}$$

Eigen-decomposition Covariance matrix



$$P(\vec{y}) = \beta e^{-\frac{1}{2}\vec{y}^T \Sigma^{-1} \vec{y}}$$

$$\Sigma^{-1} = \Phi \Lambda^{-1} \Phi^T$$

$$\Sigma = \Phi \Lambda \Phi^T$$

- Thus, our covariance matrix is just a transformation matrix that turns an isotropic Gaussian distribution (of variance 1) into non-isotropic multivariate Gaussian.

- The eigenvectors of the covariance matrix are just the basis vectors of the rotated transformation.

- And the eigenvalues of the covariance matrix are are the variances of the Gaussian in the directions of these basis vectors.

Computing a covariance matrix from data

In class Activity

Colab notebook

Make sure you download it and then upload

Thank You

Reference:

MIT- 940 Introduction to Neural Computation